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COURSE CODE: ECA0910
COURSE NAME: DIGITAL SIGNAL PROCESSING
PRACTICAL LAB RESULTS

```
EXP: 4.1:
CODE:
clc;

clear;

close all;

%% --- Input Parameters ---

fp = 1000; % Passband cutoff (Hz)
fs = 4000; % Sampling frequency (Hz)
N = 50; % Filter order (even number preferred)

wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)

%% --- 1. FIR Low Pass using Rectangular Window ---
b_rect = fir1(N, wc, 'low', rectwin(N+1));
disp('--- Rectangular Window Coefficients ---');
disp(b_rect);

%% --- 2. FIR Low Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'low', hamming(N+1));
disp('--- Hamming Window Coefficients ---');
disp(b_hamm);

%% --- 3. FIR Low Pass using Hanning Window ---
b_hann = fir1(N, wc, 'low', hanning(N+1));
disp('--- Hanning Window Coefficients ---');
disp(b_hann);

%% --- Plot Frequency Responses ---

figure;

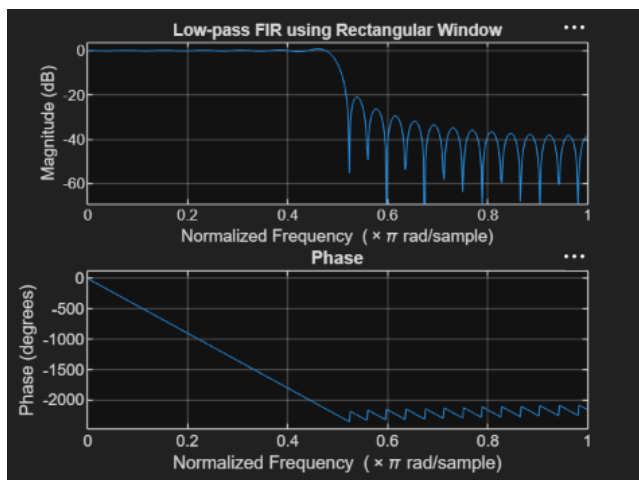
freqz(b_rect,1);
```

```

title('Low-pass FIR using Rectangular Window');
figure;
freqz(b_hamm,1);
title('Low-pass FIR using Hamming Window');
figure;
freqz(b_hann,1);
title('Low-pass FIR using Hanning Window');

```

OUTPUT:



```

--- Rectangular Window Coefficients ---
0.0126
0
-0.0137
0
0.0150
0
-0.0166
0
0.0185
0
-0.0210
0
0.0242
0
-0.0286
0
0.0349
0
-0.0449
0

```

EXP:4.2:

CODE:

```
clc;
```

```
clear;
```

```
close all;
```

```
%% --- Input Parameters ---
```

```
fp = 1000; % Passband cutoff (Hz)
```

```
fs = 4000; % Sampling frequency (Hz)
```

```
N = 50; % Filter order (even number preferred)
```

```
wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)
```

```
%% --- 1. FIR Low Pass using Rectangular Window ---
```

```
b_rect = fir1(N, wc, 'low', rectwin(N+1));
```

```
disp('--- Rectangular Window Coefficients ---');
```

```
disp(b_rect');
```

```
%% --- 2. FIR Low Pass using Hamming Window ---
```

```
b_hamm = fir1(N, wc, 'low', hamming(N+1));
```

```
disp('--- Hamming Window Coefficients ---');
```

```
disp(b_hamm');
```

```
%% --- 3. FIR Low Pass using Hanning Window ---
```

```
b_hann = fir1(N, wc, 'low', hanning(N+1));
```

```
disp('--- Hanning Window Coefficients ---');
```

```
disp(b_hann');
```

```
%% --- Plot Frequency Responses ---
```

```
figure;
```

```
freqz(b_rect,1);
```

```
title('Low-pass FIR using Rectangular Window');
```

```
figure;
```

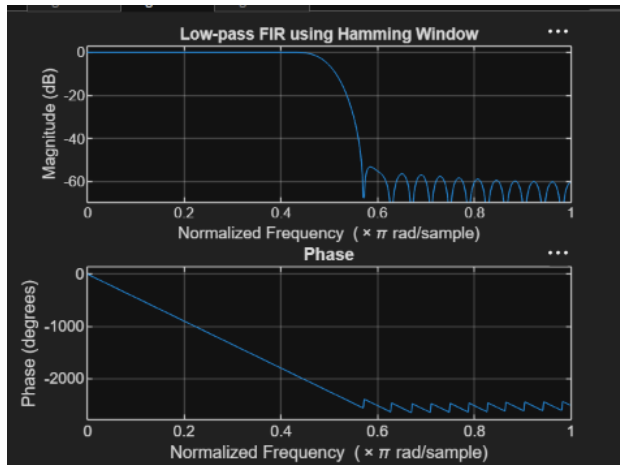
```
freqz(b_hamm,1);
```

```
title('Low-pass FIR using Hamming Window');
```

```
figure;
```

```
freqz(b_hann,1);
title('Low-pass FIR using Hanning Window');
```

OUTPUT:



```
--- Hamming Window Coefficients ---
0.0010
0
-0.0013
0
0.0021
0
-0.0034
0
0.0055
0
-0.0084
0
0.0125
0
-0.0181
0
0.0260
0
-0.0379
```

EXP:4.3:

CODE:

```
clc;
clear;
close all;

%% --- Input Parameters ---
fp = 1000; % Passband cutoff (Hz)
fs = 4000; % Sampling frequency (Hz)
N = 50; % Filter order (even number preferred)

wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)

%% --- 1. FIR Low Pass using Rectangular Window ---
b_rect = fir1(N, wc, 'low', rectwin(N+1));
disp('--- Rectangular Window Coefficients ---');
disp(b_rect);

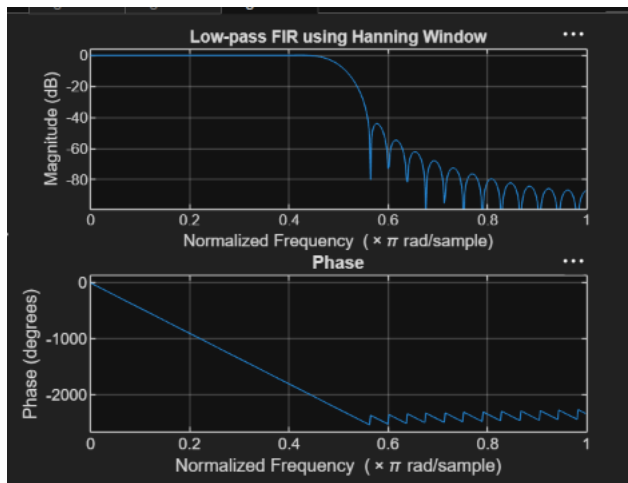
%% --- 2. FIR Low Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'low', hamming(N+1));
disp('--- Hamming Window Coefficients ---');
```

```

disp(b_hamm');
%% --- 3. FIR Low Pass using Hanning Window ---
b_hann = fir1(N, wc, 'low', hanning(N+1));
disp('--- Hanning Window Coefficients ---');
disp(b_hann');
%% --- Plot Frequency Responses ---
figure;
freqz(b_rect,1);
title('Low-pass FIR using Rectangular Window');
figure;
freqz(b_hamm,1);
title('Low-pass FIR using Hamming Window');
figure;
freqz(b_hann,1);
title('Low-pass FIR using Hanning Window');

```

OUTPUT:



```

--- Hanning Window Coefficients ---
0.0000
0
-0.0004
0
0.0013
0
-0.0028
0
0.0050
0
-0.0081
0
0.0122
0
-0.0179
0
0.0259
0
-0.0378

```